

# ELI — Architecture Ascii

ELI v2.5.3

October 2026

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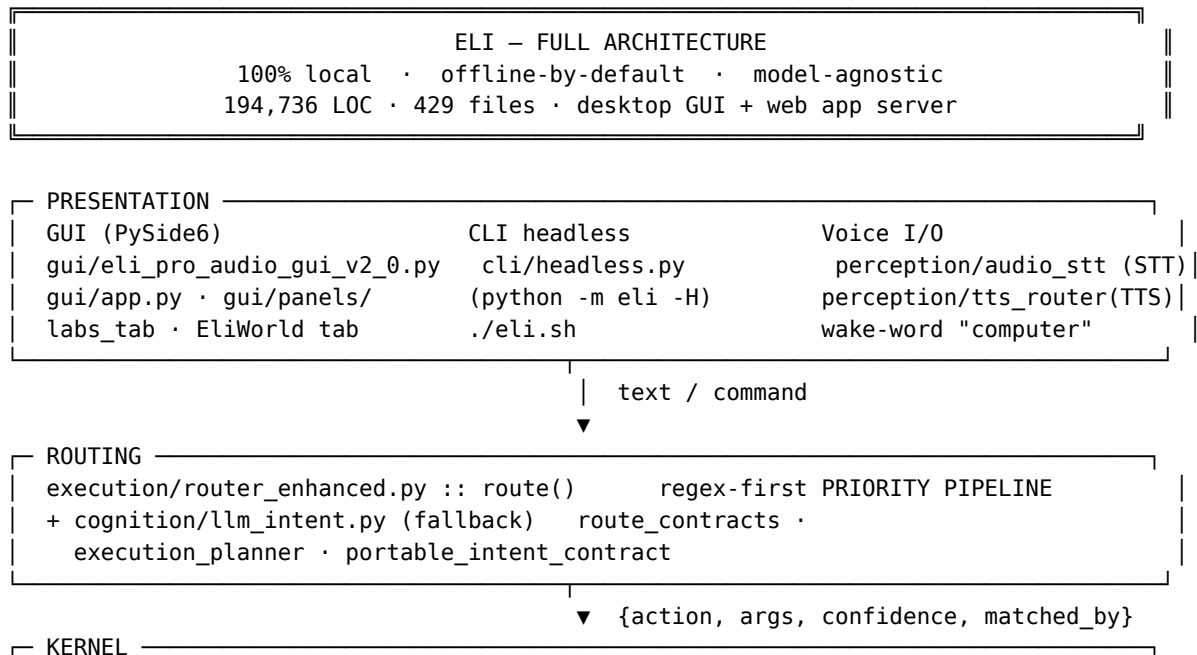
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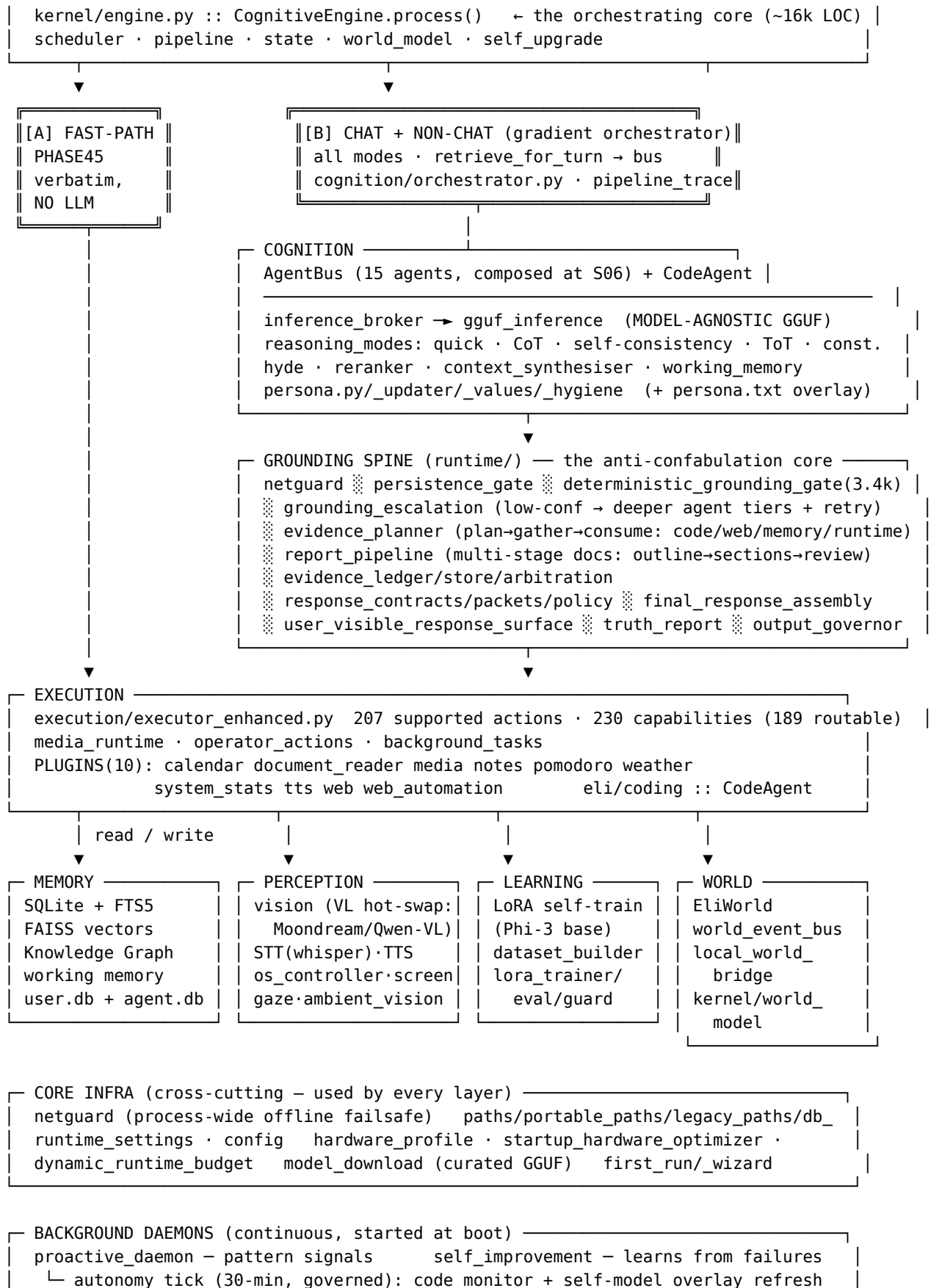
## Blueprint — ELI Full Architecture (ASCII)

**Updated for v2.4.84.** All CHAT modes → gradient orchestrator; bus composed at S06.

The entire system in one drawing, plus the module tree and data layout. Grounded in the real source (see `architecture.md` for prose, `diagrams.md` for the pipeline/memory/gating close-ups). Every layer and box maps to a real path.

### A. Full system — layered block diagram







- └─ media\_runtime.py
- └─ portable\_intent\_contract.py
- ─ cognition/            20.8k — think
  - └─ agent\_bus.py    3.5k    15 agents + dispatch
  - └─ orchestrator.py            12-stage deep retrieval
  - └─ gguf\_inference.py 3.2k · inference\_broker.py    model-agnostic inference
  - └─ reasoning\_modes.py · hyde.py · reranker.py · llm\_intent.py
  - └─ persona.py/\_updater/\_values/\_status/\_hygiene (+ persona.txt, persona.auto.txt)
  - └─ context\_synthesiser.py · working\_memory.py
  - └─ output\_governor.py · response\_governance.py · response\_sanitizer.py
  - └─ grounded\_status.py · introspection\_agent.py · tone\_analyzer.py
- ─ runtime/            35.4k — grounding spine + daemons (95 files)
  - └─ deterministic\_grounding\_gate.py 3.4k · grounding\_escalation.py
  - └─ evidence\_ledger/arbitration.py
  - └─ persistence\_gate.py · truth\_report.py · control\_contracts.py
  - └─ response\_contracts/packets/policy.py · final\_response\_assembly/provider.py
  - └─ user\_visible\_response\_surface.py · personal\_memory\*.py · reflection.py
  - └─ background\_tasks.py · self\_improvement.py · code\_monitor.py
  - └─ capability\_sync.py · pending\_proposal.py · runtime\_policy.py
  - └─ agenda.py (calendar + reminders)
- ─ memory/            8.8k — remember (11 files)
  - └─ memory.py    5.7k    Memory · SQLite + FTS5 · storage policy in policy.py
  - └─ vector\_store.py            FAISS index
- ─ perception/        10.2k — sense (23 files)
  - └─ vision.py · analyze\_image/csv/pdfs/mesh.py · ambient\_vision.py
  - └─ audio\_stt.py · local\_whisper\_stt.py · voice\_worker\_streaming.py
  - └─ tts\_router.py · os\_controller.py · screen\_locator.py · gaze\_engine.py
- ─ planning/            4.0k — proactivity (19 files)
  - └─ proactive\_daemon.py · habits\_scheduler.py · habits.py · jobqueue\_cli.py
- ─ coding/            2.1k — CodeAgent (plan→search→verify→repair, 12 files)
- ─ learning/            4.3k — LoRA self-training (14 files)
  - └─ lora\_trainer/eval/guard.py · dataset\_builder/filters.py
  - └─ bootstrap\_phi3\_base.py · base\_model\_resolver.py · training\_preflight.py
- ─ plugins/            5.9k — manager + bundled plugins (34 files)
  - └─ calendar · document\_reader · media · notes · pomodoro · weather · system\_stats · tts · web · web\_automation
- ─ world/            1.8k — EliWorld (world\_event\_bus, local\_world\_bridge)
- ─ core/            11.2k — infra
  - └─ netguard.py            offline failsafe + allow\_network()
  - └─ paths.py · portable\_paths.py · legacy\_paths.py · db\_paths.py
  - └─ runtime\_settings.py · config.py · grounding.py
  - └─ hardware\_profile.py · startup\_hardware\_optimizer.py · dynamic\_runtime\_budget.py
  - └─ model\_download.py
- ─ gui/            27.4k — PySide6 desktop

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|   |— eli_pro_audio_gui_v2_0.py 13.1k · app.py · labs_tab.py 5.7k
|   |— panels/ (startup.py: model picker + FirstBootWizard, HardwareTuningDock)
|
|— tools/ 7.6k — image_engine · news · document tools
|— contracts/ 0.8k · cli/ 0.1k · system/ 1.2k · utils/ 1.8k · setup/ 1.9k · onboarding/ 0.7k

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## D. On-disk data layout

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artifacts/
|— db/
|   |— user.sqlite3 memories · memories_archive · memory_meta · semantic ·
|   |               conversation_turns · conversations · session_summaries ·
|   |               kg_entities · kg_relations · recall_log · runtime_events ·
|   |               learning_replay · observations · habits/habit_events/habit_rules ·
|   |               user_patterns · user_model · eli_stances · belief_revisions ·
|   |               news_articles · news_reflections (+ FTS5 indexes)
|   |— agent.sqlite3 agent_dispatches · agent_metrics · improvements · failures ·
|   |               corrections · capability_proposals · error_tracking
|   |— system_index.sqlite3 OS index
|   |— coding_memory.sqlite3 coding_bug_fixes
|— vectors/index.faiss semantic index (embedder: nomic-embed-...Q4_K_M.gguf)
|— conversations/
|— runtime/ · runtime_snapshot.json live model/runtime truth
|— world/{snapshots,ledger,journal}/
|— documents/ · scripts/ · analyze_image_*/
|
config/ settings.json (gitignored) · templates/settings.template.json ·
        settings.example.json · plugins_state.json
models/ <your>.gguf · embeddings/ · whisper/ · image/ tts_piper/ Piper voices

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## E. The flow in one line

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INPUT → ROUTER → ENGINE — FAST-PATH —————→ OUTPUT (verbatim)
                        |
                        |— ORCHESTRATOR (all CHAT modes, gradient)
                        |   |— retrieve_for_turn() → MEMORY
                        |   |— dispatch_specialists() → 15-agent BUS
                        |   |— broker.infer() → finalize_turn() (S12)
GATES: netguard · persistence · grounding · escalation · governor
INFERENCE (broker → gguf, model-agnostic) → OUTPUT → TTS/GUI
BACKGROUND: proactive · self-improve · habits · learning(LoRA) · world · scheduler

Companion docs: architecture.md (prose, every subsystem) · diagrams.md (pipeline / memory
/ gating close-ups) · tools/eval/README.md (the measurement layer). “”

```